

CSCE 628 COMPUTATIONAL BIOLOGY · MODULE 10 · LECTURE NOTES

Expression and Multiple Testing

Where the computation is easy and the inference is where everything goes wrong.

MODULE OUTCOMES

- Explain what expression data is and why normalisation matters.
- Explain the count model and the small-sample problem.
- Apply multiple testing correction and explain the choice.
- Explain batch effects and how to detect them.
- Analyse an expression dataset defensibly.

1 The data

A COUNT DEPENDS ON EXPRESSION, ON TRANSCRIPT LENGTH, AND ON HOW MUCH WAS SEQUENCED

Twice as many reads assigned to a gene may mean twice the expression, a transcript twice as long, or a library that was sequenced twice as deeply — and the raw number does not distinguish between them, which makes it uninterpretable on its own.

So normalisation by library size is mandatory, and normalisation by transcript length is needed when comparing different genes within one sample — but not when comparing the same gene across samples, where the length is constant and dividing by it only adds noise. That distinction is worth getting right and is got wrong often.

And composition matters: if one gene dominates a sample, every other gene's *share* of the reads falls even though nothing about them changed — so simple proportions mislead, and the robust methods estimate a per-sample size factor from the bulk of genes instead.

Which is CSCE 679 Module 08 §3's normalisation argument — a map of counts is a map of population — arriving in a field where the denominator is itself estimated from the data rather than known.

2 The model

- Counts are modelled as negative binomial — Poisson counting noise plus extra variance — because biological replicates vary considerably more than counting noise alone would predict, and a Poisson model therefore declares far too much significant.

- **And the per-gene variance must be estimated from perhaps three replicates, which is hopeless on its own** — so **the standard methods share information across genes**, fitting a mean-variance trend and shrinking each gene's estimate toward it.
- **Which is the key statistical idea in this field — borrowing strength across thousands of parallel tests — and it works well and is an assumption:** that genes with similar expression have similar variance, which is usually but not always true.
- **Then a test per gene, and a p-value per gene — twenty thousand of them, which is §3's subject** and is where the list becomes a finding or does not.
- **And the effect size matters as much as the p-value: a tiny fold change can be highly significant given enough samples, and statistical significance is not biological importance** — which is why the standard practice reports both and filters on both.

3 Multiple testing

```
THE ARITHMETIC
20,000 tests at alpha = 0.05
=> 1,000 expected false positives
if you report 1,200 "significant" genes, most of
your list is noise.
```

```
BONFERRONI
use alpha / m. Controls the probability of ANY
false positive (the family-wise error rate).
Correct, and far too strict here: you will find
almost nothing, and the true effects are small.
```

```
BENJAMINI-HOCHBERG
controls the FALSE DISCOVERY RATE: the expected
PROPORTION of your reported list that is false.
q = 0.05 means about 5% of your hits are
expected to be wrong.
Which is the right target for a screen whose
output is a list for experimental follow-up.
```

```
AND THE COUNT OF TESTS MUST INCLUDE THE ONES YOU RAN
AND DISCARDED. Everybody omits those.
```

The count of tests must include the ones you ran and discarded — the normalisations you tried, the covariate sets you fitted, the subgroups you looked at — **which everybody omits, and which is the honest part of the correction. FDR for screens, FWER when a single false positive is costly**: the choice follows from what the list is for, and should be stated along with the number m (CSCE 676 Module 11's multiplicity material, which is the general treatment).

4 Batch effects

IF YOUR CASES AND CONTROLS WERE PROCESSED ON DIFFERENT DAYS, YOU CANNOT SEPARATE BIOLOGY FROM BATCH

Processing date, technician, reagent lot, instrument, and position on a plate all produce systematic measurable differences — frequently larger than the biological effect you are looking for, which is the uncomfortable part.

And if batch is confounded with the condition — all cases run in March, all controls in April — then no statistical method can separate them: the information is simply not in the data, which is Module 05 §3's kind of limit arriving in a statistical setting rather than a combinatorial one.

So the fix is in the experimental design: randomise or deliberately balance the samples across batches, and record the batch metadata — which is free and is very often not done, leaving the analyst with an unanswerable question years later.

And the detection is cheap: cluster or project the samples and colour the points by batch — if they separate by batch rather than by condition, you have your answer before running a single test (CSCE 679 Module 07 §2's projection caveats apply, but the gross signal is unmistakable).

AND THE GENERAL SHAPE OF THIS MODULE

The computation here is trivial and the inference is where everything fails — which is the exact opposite of Modules 02 through 09, and is why this module is placed after them rather than before.

And the failures are systematic and named: unnormalised counts, uncorrected multiple testing, and batch confounded with condition — three errors that account for a large share of the irreproducible findings in this literature (Module 01's closing callout).

All three are detectable in an afternoon by somebody who knows to look — which is the practical value of this module and the reason it is worth more than its algorithmic content suggests.

So: normalise, correct, and plot the samples coloured by batch before anything else — three steps, and Project 2 requires all three, with the batch structure either examined or stated as unexaminable from the available metadata.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Love, Huber & Anders — Moderated estimation of fold change and dispersion (DESeq2) (free)	§§1 and 2 — the shrinkage model, explained by its authors, and the software is the standard.
Benjamini & Hochberg — Controlling the false discovery rate	§3 — the original, and the idea is simpler than its reputation.

Resource	What it covers for this module
Leek et al. — Tackling the widespread and critical impact of batch effects (free)	§4 — with worked examples of published findings that were batch structure.
Conesa et al. — A survey of best practices for RNA-seq data analysis (free)	The whole module , as a practical checklist.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Take raw counts** and compute three normalisations; compare the gene rankings.
- 2 **Show that length normalisation is wrong** for a within-gene-across-sample comparison.
- 3 **Simulate a dominant gene** and show that proportions mislead.
- 4 **Fit a Poisson and a negative binomial** to the same counts and compare the significance.
- 5 **Estimate per-gene variance from $n = 3$** and compare to the shrunk estimate.
- 6 **Run 20,000 tests on pure noise** and count the p-values below 0.05.
- 7 **Apply Bonferroni and BH** to the same p-values and compare the lists.
- 8 **Count the tests you actually ran** in one analysis, including discarded ones.
- 9 **Cluster samples and colour by batch** on a public dataset.
- 10 **Find a published dataset** whose batch metadata is not available.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Name three things a raw count depends on.
- 2 When is length normalisation right, and when wrong?
- 3 Why do proportions mislead, and what is used instead?
- 4 Why negative binomial rather than Poisson?
- 5 What is shrinkage doing, and what does it assume?
- 6 Why is effect size needed alongside the p-value?
- 7 Give the arithmetic of 20,000 tests.
- 8 Contrast FWER and FDR, and say when each is right.
- 9 Which tests must be counted?

10 Why can confounded batch not be corrected, and how is it detected?